EDGEVISION

V3-HN PERFORMANCE BENCHMARK

Technical System Throughput Document

1. Benchmark Environment

The benchmark results in this document reflect system performance under real-time video processing conditions. The reported 16.2 FPS metric was obtained on an RTX development workstation running PyTorch with FP16 precision. This does NOT represent Jetson edge performance.

2. Performance Table

Throughput metrics for the video processing pipeline:

Metric	V2 Baseline	V3-HN Production
Warm FPS	24.3 FPS	16.2 FPS
Minimum Requirement	12.0 FPS	12.0 FPS
Status	PASS	PASS

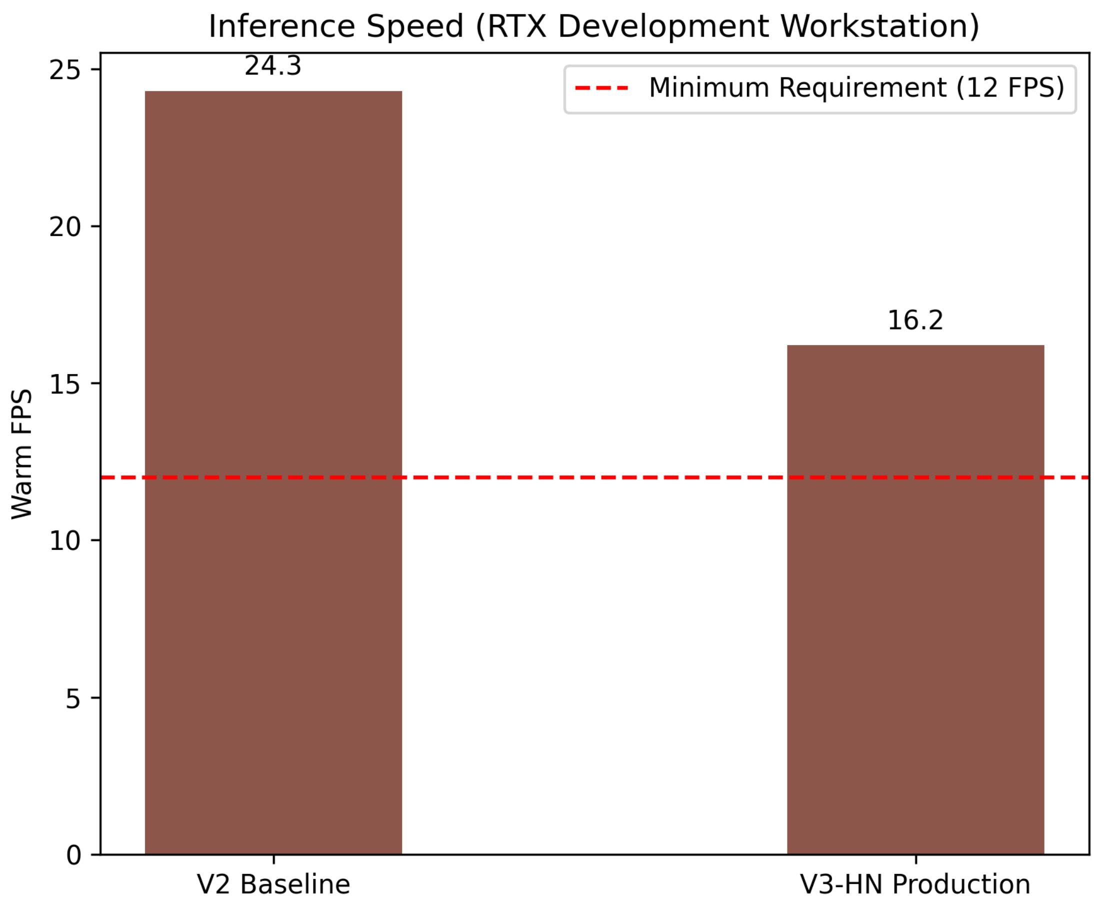

Figure: FPS Comparison Against Required Minimum

3. Latency

End-to-end P95 Latency:

- V3-HN P95 Latency: 134.63 ms

- V2

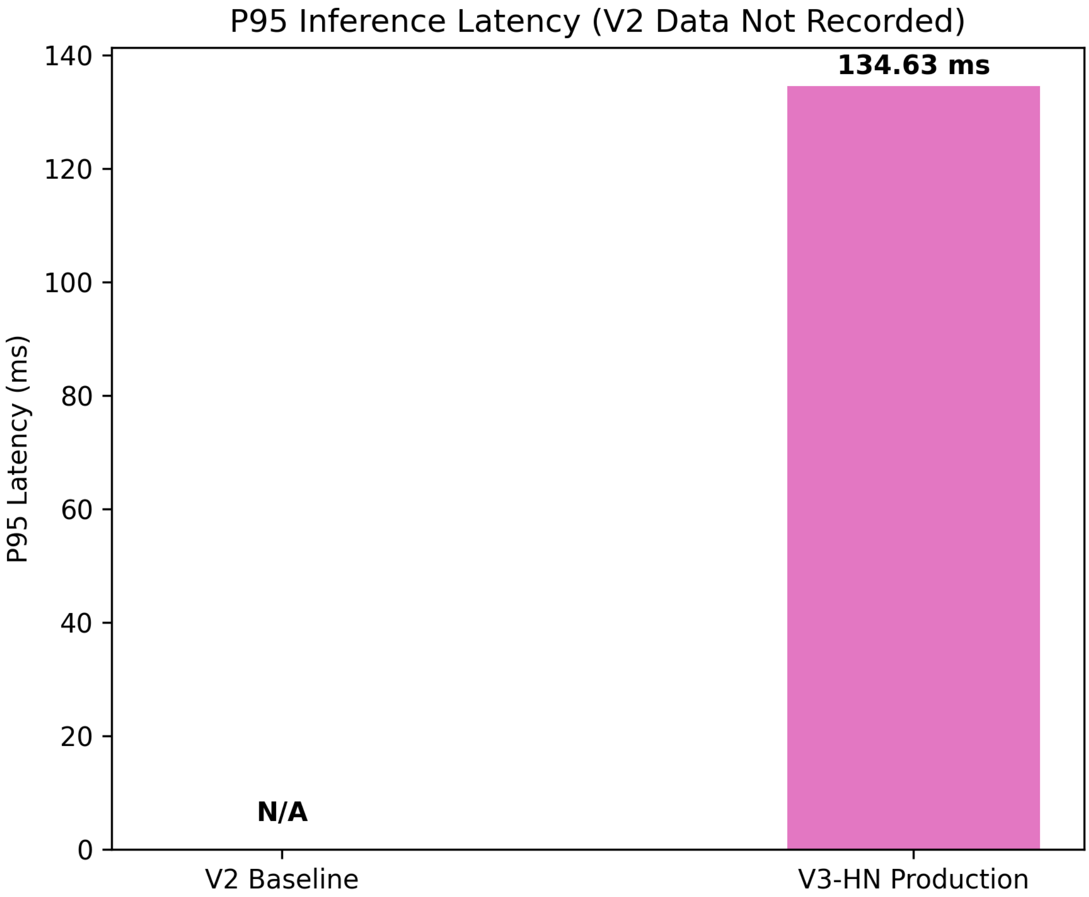

Figure: P95 Inference Latency

4. Real-World Functional Benchmark

Functional reliability of the pipeline in real-world scenarios:

Validation Metric	V2 Baseline	V3-HN Production
Helmet False Positives	154	0
Vest False Positives	10	0
Confirmed Violations	14	14

Pipeline Validation Checks:

- Association Failures: 0

- Dup

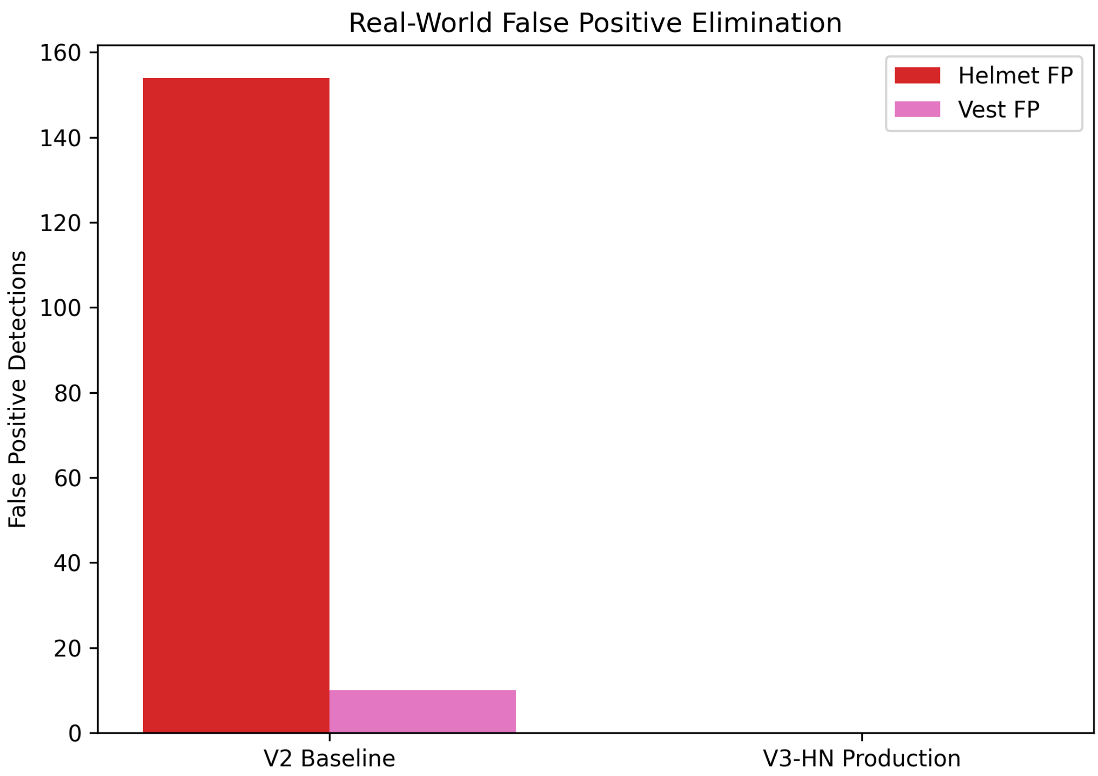

Figure: False Positive Reliability Check

5. Hardware & Deployment Status

DEVELOPMENT WORKSTATION:

- Measured FPS: 16.2

- Sta

TARGET JETSON EDGE HARDWARE:

- Hardware Validation: PENDING HARDWARE VALIDATION

The development workstation FPS proves software viability, but does NOT prove Jetson physical performance.

Deployment Status:

- ONNX Export Script: READY

- Ter

6. Conclusion

The EdgeVision V3-HN pipeline satisfies the current software/development benchmark requirement of 12 FPS, sustaining 16.2 FPS on the development workstation with reliable functional detection limits. Physical Jetson validation remains pending hardware availability.